

ELEC 473 – Cryptography and Network Security

Quiz 2

Monday, Nov. 8, 2021, 11:30-12:15 PM

INSTRUCTIONS—Read Carefully

- Write your name and ID legibly at the top of the papers.
- Calculators are allowed to use.
- Show all your work. Partial credit will be given for substantial progress towards the solution. No credit will be given for answers with no substantial process.

Q1(10 Marks) Solving the following system of congruences:

$$x \equiv 5 \pmod{7}$$

$$x \equiv 3 \pmod{11}$$

$$x \equiv 5 \pmod{13}$$

$$(1') \begin{cases} \gcd(7, 11) = 1 \\ \gcd(7, 13) = 1 \\ \gcd(11, 13) = 1 \end{cases}$$

$$M = m_1 m_2 m_3 = 7 \times 11 \times 13 = 1001$$

$$M_1 = m_2 \cdot m_3 = 11 \times 13 = 143 \quad (1')$$

$$M_2 = m_1 \cdot m_3 = 7 \times 13 = 91 \quad (1')$$

$$M_3 = m_1 \cdot m_2 = 7 \times 11 = 77 \quad (1'')$$

$$y_1 = M_1^{-1} \pmod{m_1} = 143^{-1} \pmod{7} = 5 \pmod{7} \quad (1')$$

$$y_2 = M_2^{-1} \pmod{m_2} = 91^{-1} \pmod{11} = 4 \pmod{11} \quad (1')$$

$$y_3 = M_3^{-1} \pmod{m_3} = 77^{-1} \pmod{13} = 12 \pmod{13} \quad (1')$$

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \pmod{M} \quad (2')$$

$$\equiv 5 \times 143 \times 5 + 3 \times 91 \times 4 + 5 \times 77 \times 12 \pmod{1001}$$

$$\equiv 278 \pmod{1001} \quad (1')$$

Q2 (10 Marks) For $n=pq$, where p and q are distinct odd primes, define

$$\lambda(n) = \frac{(p-1)(q-1)}{\gcd(p-1, q-1)}.$$

Suppose that we modify the RSA cryptosystem by requiring that $ed \equiv 1 \pmod{\lambda(n)}$. The public key is (n, e) and the private key is d .

a. Prove that encryption and decryption of any plaintext $m \in \mathbb{Z}_n^*$ are still the inverse operations in this modified cryptosystem.

b. If $p=17$, $q=41$, and $e=7$, compute d in this modified cryptosystem.

$$\begin{aligned} \text{a. } ed \equiv 1 \pmod{\lambda(n)} &\Rightarrow ed = k\lambda(n) + 1, k \in \mathbb{Z} \\ \text{if } \gcd(m, n) = 1 &\Rightarrow \gcd(m, p) = 1, \quad m^{\lambda(n)} \equiv m^{\frac{(p-1)(q-1)}{\gcd(p-1, q-1)}} \equiv m^{\frac{(p-1)(q-1)}{1}} \equiv 1 \pmod{p} \\ &\quad \gcd(m, q) = 1, \quad m^{\lambda(n)} \equiv m^{\frac{(p-1)(q-1)}{\gcd(p-1, q-1)}} \equiv m^{\frac{(p-1)(q-1)}{1}} \equiv 1 \pmod{q}. \end{aligned}$$

$$\begin{aligned} \text{So any } m \in \mathbb{Z}_n^* &\begin{cases} m^{\lambda(n)} \equiv 1 \pmod{p} \\ m^{\lambda(n)} \equiv 1 \pmod{q} \end{cases} \xrightarrow{\text{CRT}} m^{\lambda(n)} \equiv 1 \pmod{n} \quad (4') \\ m^{ed} &\equiv m^{k\lambda(n)+1} \equiv [m^{\lambda(n)}]^k \cdot m \equiv m \pmod{n} \quad (2') \quad \text{(Use Euler's Theorem or Fermat's Little Theorem)} \end{aligned}$$

$$\text{So encryption and decryption are still inverse operations}$$

$$\text{b. } \lambda(n) = \frac{(p-1)(q-1)}{\gcd(p-1, q-1)} = \frac{(17-1)(41-1)}{\gcd(16, 40)} = \frac{16 \times 40}{8} = 80 \quad (2')$$

$$d = e^{-1} \pmod{\lambda(n)} = 7^{-1} \pmod{80} = 23 \quad (2')$$

Q3 (10 Marks) Let E denote the elliptic curve $y^2 = x^3 + x + 6$, defined over Z_{11} . It can be shown that $\#E=13$. Consider the Elliptic Curve Digital Signature Algorithm (ECDSA) presented in Cryptosystem 1. Suppose the ECDSA is implemented in E , with $A=(8, 3)$, $m=5$.

- Compute the public verification key $B=mA$.
- Compute the signature on a message x if $\text{SHA3-224}(x)=3$, when $k=7$.
- Show the computations used to verify the signature constructed in part b.

(Hint: $P=(2,7)$, $2P=(5,2)$, $3P=(8,3)$, $4P=(10,2)$, $5P=(3,6)$, $6P=(7,9)$, $7P=(7,2)$, $8P=(3,5)$, $9P=(10,9)$, $10P=(8,8)$, $11P=(5,9)$, $12P=(2,4)$)

Cryptosystem 1. Elliptic Curve Digital Signature Algorithm (ECDSA)

KeyGen. Let p be a large prime and let E be an elliptic curve defined over Z_p . Let A be a point on E having prime order q , such that the Discrete Logarithm problem in $\langle A \rangle$ is infeasible, where $\langle A \rangle$ denotes the group generated by the generator A . Let $A = Z_q^* \times Z_q^*$, and define

$$K = \{(p, q, E, A, m, B) : B = mA\},$$

where $0 \leq m \leq q-1$. p, q, E, A , and B are the public verification key, and m is the secret signing key.

Sign. For $K = (p, q, E, A, m, B)$, a message $x \in \{0, 1\}^*$, and a (secret) random number k , $1 \leq k \leq q-1$, define

$$\sigma = \text{Sign}(K, x, k) = (r, s),$$

where $kA = (u, v)$

$$r = u \bmod q, \text{ and}$$

$$s = k^{-1} (\text{SHA3-224}(x) + mr) \bmod q.$$

(If either $r = 0$ or $s = 0$, a new random value of k should be chosen.)

Verif. For the message x and the signature $\sigma = (r, s)$, verification is done by performing the following computations:

$$w = s^{-1} \bmod q$$

$$i = w \times \text{SHA3-224}(x) \bmod q$$

$$j = wr \bmod q$$

$$(u, v) = iA + jB$$

$$\text{Verif}(x, \sigma, B) = \text{true} \Leftrightarrow u \bmod q = r.$$

a. $B = mA = 5(8, 3) = 5 \times 3P = 15P = 2P = (5, 2) \bmod 13$ (3')

b. $(u, v) = kA = 7(8, 3) = 7 \times 3P = 21P = 8P = (3, 5) \bmod 13$

(1') $u = 3 \bmod 13, \quad v = 5 \bmod 13$

(1') $r = u \bmod q = 3 \bmod 13$

(1') $s = k^{-1} (\text{SHA3-224}(x) + mr) \bmod q = 7^{-1} (3 + 5 \times 3) \bmod 13 = 2 \times 5 \bmod 13 = 10 \bmod 13$

$\sigma = (3, 10)$

(1') c. $w = s^{-1} \bmod q = 10^{-1} \bmod 13 = 4 \bmod 13$

(1') $i = 4 \times 3 \bmod 13 = 12 \bmod 13$

(1') $j = 4 \times 3 \bmod 13 = 12 \bmod 13$

(1') $(u, v) = iA + jB = 12(8, 3) + 12(5, 2) = 12 \times 3P + 12 \times 2P = 8P = (3, 5) \bmod 13$
 $u = 3 \bmod 13 = r \Rightarrow \text{Verif}(x, \sigma, B) = \text{true}$